E6. LA REAZIONE A CATENA DELLA POLIMERASI
(*Polymerase Chain Reaction*, PCR; Saiki *et al.*, 1988)

E6.1 The standard PCR reaction

PCR amplification involves **two oligonucleotide primers that flank the DNA segment to be amplified** and repeated cycles of heat denaturation of the DNA, annealing of the primers to their complementary sequences, and extension of the annealed primers with DNA polymerase. These primers hybridize to opposite strands of the target sequence and are oriented so DNA synthesis by the polymerase proceeds across the region between the primers (Figure E6.1), effectively doubling the amount of that DNA segment.

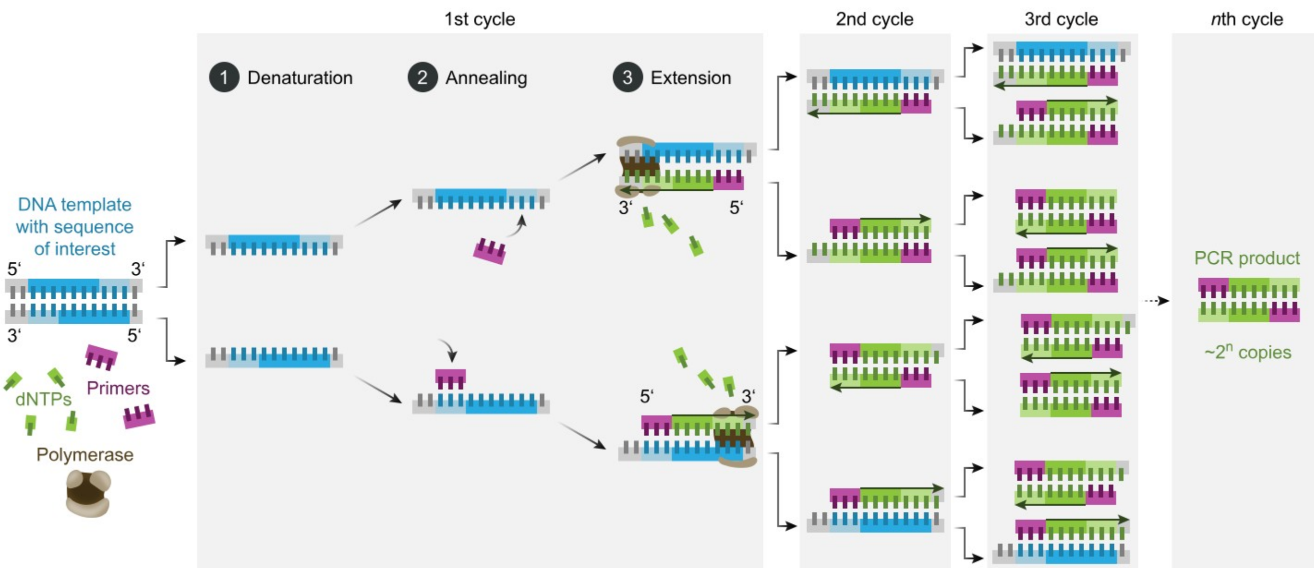

Figure E6.1 Schematic drawing of a complete PCR cycle.

Moreover, since the extension products are also complementary to and capable of binding primers, each successive cycle essentially doubles the amount of DNA synthesized in the previous cycle. This results in the exponential accumulation of the specific target fragment, approximately 2^n, where n is the number of cycles.

One of the drawbacks of the method, however, is the thermolability of the Klenow fragment of *Escherichia coli* DNA polymerase I used to catalyze the extension of the annealed primers. Because of the heat denaturation step required to separate the newly synthesized strands of DNA, fresh enzyme must be added during each cycle – a tedious and error-prone process if several samples are amplified simultaneously. We now describe the replacement of the *E. coli* DNA polymerase with a **thermostable DNA polymerase purified from the thermophilic bacterium, *Thermus aquaticus* (Taq)**, that can survive extended incubation at 95°C. Since this heat-resistant polymerase is relatively unaffected by the denaturation step, it does not need to be replenished at each cycle. This modification not only simplifies the procedure, making it amenable to automation, it also substantially improves the overall

performance of the reaction by increasing the specificity, yield, sensitivity, and length of targets that can be amplified.
Samples of human genomic DNA were subjected to 20 to 35 cycles of PCR amplification with optimal amounts of either Klenow or Taq DNA polymerase and analyzed by agarose gel electrophoresis (Figure 2.2A) and Southern hybridization (Figure E6.2B).

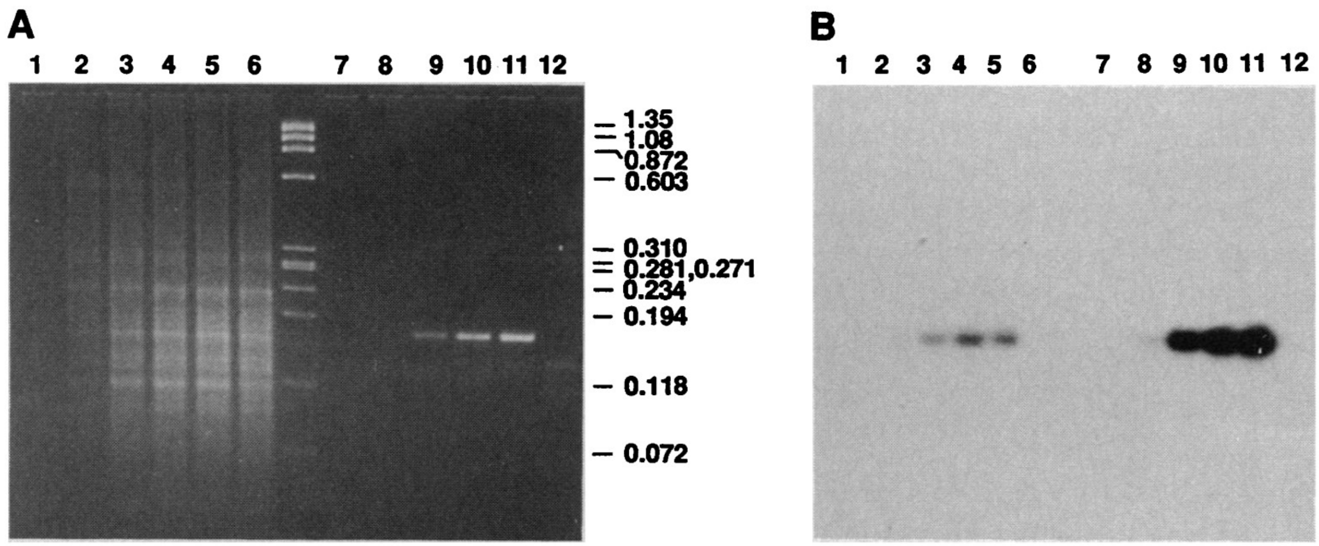

Figure E6.2 Comparison of Klenow and Taq DNA polymerase-catalyzed PCR amplification products of the human β-globin gene.
(A) Electrophoretic analysis of the PCR products obtained with Klenow polymerase (lanes 1 to 6) and Taq polymerase (lanes 7 to 12) after 0 cycles (lanes 1 and 7), 20 cycles (lanes 2 and 8), 25 cycles (lanes 3 and 9), **30 cycles** (**lanes 4 and 10**), and 35 cycles (lanes 5, 6, 11, and 12) of amplification. The DNA samples that were amplified were prepared from the human cell lines MOLT4 (lanes 1 to 5 and 7 to 11) and GM2064 (lanes 6 and 12). MOLT4 is homozygous for the normal 3-globin gene, GM2064 possesses a homozygous deletion of the entire 3-globin gene complex. The molecular size marker is 250 ng of Hae HI-digested 4)X174 replicative form (RF) DNA (New England Biolabs).
(B) Southem analysis of the gel with a 32P-labeled oligonudceotide probe. Compared against standards, the intensities of the bands in the Kienow amplifications were significantly increased.
Amplifications with Taq polymerase took place in 100 µl reaction mixtures containing 1 µg of genomic DNA in 50 mM KCl, 10 mM tris (pH 8.4), 2.5 mM $MgCl_2$, each primer (PCO3 and KM38) at 1 µM, each dNTP (dATP, dCTP, dTTP, dGTP) at 200 µM, gelatin at 200 µg/ml, and 2 units of polymerase. The samples were overlaid with several drops (~ 100 µl) of mineral oil to prevent condensation and subjected to 20 to 35 cycles of amplification as follows. The samples were heated from 70° to 95°C over a 1 minute period (to denature the DNA), cooled to 40°C over 2 minutes (to anneal the primers), heated to 70°C in 1 minute (to "activate" the polymerase), and incubated at that temperature for **0.5 minute** (to extend the annealed primers). Additional Taq DNA polymerase was not added to the samples during amplification. **One unit of enzyme is the amount that will incorporate 10 nmol of total deoxyribonudeotide triphosphates into acid-precipitable material in 30 minutes at 74°C with activated salmon sperm DNA as template**. After the last cycle, all samples were incubated for an additional 5 to 10 minutes at 37° or 70°C to ensure that the final extension step was complete. After precipitation with ethanol and resuspension in 100 µl TE buffer, each sample (8 µl) was resolved on a composite gel of 3% NuSieve and 1% SeaKem agaroses (FMC) in tris-borate buffer and stained with ethidium bromide. Southern transfers were performed onto Genetrans45 nylon membranes. A 19-base oligonucdeotide probe specific for the amplified gobin fagmnent, 19A, was 5' end-labeled with 32p and hybridized to the filter. The autoradiogram was exposed for 3 hours with a single intensification screen.

The PCR primers direct the synthesis of a 167-bp segment of the human β-globin gene. Electrophoretic examination of the reactions catalyzed by the Klenow polymerase reveals a **broad molecular size distribution of amplification products that is presumably the result of nonspecific annealing and extension of primers to unrelated genomic sequences under what are essentially nonstringent hybridization conditions**: Klenow polymerase reaction buffer at 37°C. Nevertheless, Southern blot analysis with a β-globin hybridization probe reveals the β-globin amplification fragment in all samples in which the β-globin target sequence was present.
A substantially different electrophoretic pattern is seen in the amplifications performed with Taq DNA polymerase, where the single predominant band is the 167 bp target segment. **This specificity is evidently due to the temperature at which the primers are extended**. Although the annealing step is performed at 40°C, the temperature of Taq polymerase-catalyzed reactions was raised to about 70°C, near the temperature optimum of the enzyme. **During the transition from 40° to 70°C, poorly matched primer-template hybrids (which formed at 40°C) dissociate**, and only highly complementary substrate remains as the reaction approaches the temperature at which catalysis occurs. Furthermore, because of increased specificity, there are fewer nonspecific extension products to compete for the polymerase, and the yields of the specific target fragment are higher.
The exponential accumulation of PCR amplification products is not an unlimited process. Eventually, **a level of amplification is reached where more primer-template substrate has accumulated than the amount of enzyme present is capable of completely extending in the allotted time**. When this occurs, the efficiency of the reaction declines, and the amount of PCR product accumulates in a linear rather than an exponential manner. The Klenow polymerase reaction begins to "plateau" around the 20th cycle, after a 3 x 10^5 fold amplification of the β-globin target sequence. The higher specificity of the Taq polymerase-catalyzed reaction, however, permits it to proceed efficiently for an additional five cycles, to an amplification level of 4 x 10^6 before the activity of the polymerase becomes limiting.
The specificity of the Taq DNA polymerase-mediated amplifications can be affected by the time allowed for the primer extension step and by the **quantity of enzyme used in the reaction**. The electrophoretic patterns of PCR products obtained with different extension times and units of Taq DNA polymerase on otherwise identical samples indicate that **the amount of nonspecific DNA increases as the extension times become longer or the number of Taq polymerase units increases**.
A significant improvement in specificity is obtained when the temperature of the primer annealing step is raised from 40° to 55°C. This effect is demonstrated by the amplification of the β-globin gene in a set of dilutions of normal genomic DNA into the DNA of a mutant cell line with a homozygous deletion of the β-globin gene. These samples, each containing 2 μg of DNA, represent β-globin gene frequencies that range from one copy per genome (two copies per diploid cell) in the undiluted normal DNA sample, to as little as one copy per 10^6 genomes (one copy per 500,000 cells) in the 10^-6 dilution. After 40 cycles of PCR with primer annealing at 40°C, the specific amplification fragment can be seen in the 10^-2 dilution by electrophoretic examination and in the 10^-4 dilution by Southem analysis. However, **in parallel samples amplified with 55°C annealing, much less nonspecific DNA is present** and the product band is visible in the original gel in the 10^-4 dilution and, by Southem analysis, in the 10^-6 dilution. Although **annealing at 55°C is usualy of limited value when amplifying genomic targets present at one or two copies per cell**, the reduction in the amount of nonspecific primer extension products improves the limit of sensitivity by two orders of magnitude.
Since 2 μg of the 10^-6 dilution would contain, on average, only 0.6 copy of the β-globin gene, these data suggest that PCR with Taq polymerase may be able to amplify a single target molecule in 10^5 to 10^6 cells.

This result was confirmed by demonstrating a Poisson distribution of successful amplifications on limiting amounts of the β-globin template. Fifteen identical 1 μg samples of the same 10^-6 dilution were amplified for 60 cydes with annealing at 55°C and analyzed for the presence of a β-globin amplification product by Southern hybridization. Each of these samples of genomic DNA (equivalent to that in 150,000 cells) should have, on average, 0.3 copy of the β-globin gene. The fraction of those samples that contain at least one copy is 4 out of 15; the remaining 11 should contain no β-globin templates. The observed frequency was 9 of 15. Although this is twice as high as the expected value, it does not significantly affect the interpretation of the results. The distribution of successful amplifications is consistent with a concentration of approximately one β-globin gene per 500,000 cells which indicates that of the nine positive samples, most, if not all, originally contained a single copy of the 3-globin gene. If only one target molecule was present initially, the intensities of the PCR products in the 10^-6 samples are equivalent to amplifications of about 10^9 to 10^10.
This experiment supports the conclusion that **Taq polymerase-mediated PCR performed with primer annealing at 55°C is able to successfully amplify a single target molecule in a DNA background of 10^5 cells**.
Subsequent investigations have demonstrated that the reaction can detect a single copy of target DNA in 10 μg of DNA, the cquivalent of 1.5 million diploid cells.

	Saiki *et al.*, 1988	Taberlet *et al.*, 2018
Volume di soluzione	100 μl	20 μl
Volume della master mix		10 μl
Acqua DNA-free		4 μl
KCl	50 mM	
Tris-HCl	10 mM	
$MgCl_2$	2.5 mM	
primer	1 μM	
forward		2 μl di una soluzione 2 μM
reverse		2 μl di una soluzione 2 μM
dNTP	200 μM	
Tipo di polimerasi	Taq	AmpliTaq Gold 360 Master Mix
Quantità di enzima	2 unità	
Quantità di DNA	1 μg	100 ng
Temperatura di denaturazione	95°C	95°C
Tempo di denaturazione	1 minuto	30 secondi
Temperatura di *annealing* del *primer*	55°C	50°C
Tempo di *annealing* del *primer*	2 minuti	30 secondi
Tempo di attivazione della polimerasi	1 minuto	10 minuti
Temperatura di attivazione della pol.	70°C	95°C
Temperatura di estensione del *primer*	70°C	72°C
Tempo di estensione del *primer*	30 secondi	1 minuto
N° di cicli	30	35

Tabella E6.1 Parametri utilizzati per la PCR in due studi diversi.

E6.2 I Microsatelliti

Se si sottopongono a migrazione elettroforetica i prodotti della digestione con endonucleasi di restrizione di un campione di DNA nucleare di dimensioni cospicue, il profilo che si ottiene è rappresentato da una strisciata: le diverse taglie dei frammenti non possono essere separate perché costituiscono un continuum, mentre le uniche bande visibili corrispondono alle cosiddette "sequenze satellite". Queste ultime furono così definite poiché, in esperimenti di ultracentrifugazione attraverso gradienti continui di CsCl (cloruro di cesio), si riscontrava la presenza di un picco "satellite" del picco principale, che fu ad esse attribuito. Le sequenze satellite, altamente ripetitive, sono organizzate in raggruppamenti (clusters) a livello centromerico e telomerico, ossia nelle regioni eterocromatiche dei cromosomi. Esistono però anche sequenze satellite disperse, in maniera apparentemente casuale, lungo tutto il genoma: secondo la lunghezza, si distinguono sequenze LINE (Long INterspersed Elements) e sequenze SINE (Short INterspersed Elements). In analogia con le sequenze satellite vere e proprie, vengono identificate altre due categorie eterogenee di sequenze ripetute, i minisatelliti ed i microsatelliti. I minisatelliti si trovano, negli organismi eucarioti, nelle regioni eucromatiche dei cromosomi e sono costituiti dalla ripetizione testa-coda (a coprire un'estensione di 0,5-3 Kb) di un modulo formato da un numero di paia di basi compreso fra 9 e 100 e in media 15 bp. I microsatelliti, invece, sono costituiti da brevi moduli1 (2-6 bp) ripetuti testa-coda un numero moderato di volte. Il genoma umano contiene almeno 30.000 loci di microsatelliti, localizzati nelle regioni eucromatiche. Minisatelliti e microsatelliti costituiscono un'importante classe di markers denominati, nel loro insieme, "Variable Number of Tandem Repeats" (numero variabile di gruppi di sequenze ripetute). I VNTR sono ipervariabili: in ogni specie animale, per ciascun *locus* VNTR, infatti, esistono numerosissimi alleli, la frequenza dei quali varia nelle diverse popolazioni. Ognuno di essi è identificato dal numero delle sequenze ripetute (moduli). Ad ogni *locus* VNTR gli alleli vengono ereditati in modo mendeliano, cioè ogni individuo possiede soltanto due alleli ad ogni locus VNTR (uno di origine materna e l'altro di origine paterna), ma il numero di alleli presenti nella popolazione è così elevato che la maggior parte degli individui è eterozigote. Dal punto di vista medico-legale (ma anche negli studi sui rapporti di parentela fra individui di popolazioni animali naturali o allevate), ciò consente sia di escludere che uno specifico soggetto sia figlio di due determinati individui (presunti genitori) sia, con un'alta probabilità, di assegnare la paternità di quel soggetto ad un'altra coppia di genitori. Inoltre, se i due alleli per locus vengono moltiplicati per le decine di migliaia di loci VNTR presenti nel genoma di un mammifero, si ottiene un numero pressoché infinito di combinazioni possibili2. Ogni singolo individuo di una determinata specie animale potrà quindi essere inequivocabilmente identificato sulla base dell'assetto di alleli, ad un certo numero di loci VNTR, di cui è portatore: tale metodo di tipizzazione genetica prende il nome di DNA-fingerprintig ("impronta digitale" genetica). L'elevato numero di *loci* microsatellitari esistenti e di alleli per *locus* possono essere sfruttati, oltre che per applicazioni diagnostiche, medico-legali e forensiche, anche per lo studio della variabilità genetica delle popolazioni animali o vegetali naturali. Dal punto di vista operativo, la tecnica della PCR può essere vantaggiosamente applicata anche per l'amplificazione dei loci VNTR, poiché questi ultimi sono in genere preceduti e seguiti da sequenze invarianti, sulle quali dovranno essere costruiti i primers. Una certa conoscenza delle sequenze che fiancheggiano i loci VNTR sarà dunque necessaria, e ciò rappresenta un'importante differenza rispetto alla tecnica RAPD, per l'applicazione della quale non è richiesta alcuna informazione preliminare sulle sequenze da amplificare o su quelle che le fiancheggiano. Cionondimeno, la disponibilità di un grande numero di primers quasi universali riduce notevolmente il lavoro preliminare e l'impegno che deve essere profuso quando si inizia a studiare una specie nuova. Riassumendo, l'amplificazione mediante PCR dei loci VNTR offre a considerare i seguenti aspetti:

- L'amplificazione di loci VNTR produce profili elettroforetici individuali;
- I diversi alleli per i singoli loci sono direttamente osservabili su di un gel di agaroso o di poliacrilammide;

- **Se le condizioni di reazione sono compatibili, numerosi *loci* possono essere co-amplificati nella stessa miscela di reazione** (multiplex PCR).

E6.2.1 I *loci* microsatelliti descritti in *Myotis myotis* da Castella e Ruedi (2000)

In R:

```
> library(ape)
> library(seqinr)
> mm = paste('AF2036', 37:77, sep='')
> mm
 [1] "AF203637" "AF203638" "AF203639" "AF203640" "AF203641" "AF203642"
 [7] "AF203643" "AF203644" "AF203645" "AF203646" "AF203647" "AF203648"
[13] "AF203649" "AF203650" "AF203651" "AF203652" "AF203653" "AF203654"
[19] "AF203655" "AF203656" "AF203657" "AF203658" "AF203659" "AF203660"
[25] "AF203661" "AF203662" "AF203663" "AF203664" "AF203665" "AF203666"
[31] "AF203667" "AF203668" "AF203669" "AF203670" "AF203671" "AF203672"
[37] "AF203673" "AF203674" "AF203675" "AF203676" "AF203677"
> myotis.seq = read.GenBank(mm, as.character = TRUE)
> dir.create('~/BATS/DNA_SEQUENCES')
> setwd('~/BATS/DNA_SEQUENCES')
> getwd()
[1] "/home/piero/BATS/DNA_SEQUENCES"
> write.fasta(sequences = myotis.seq, names = names(myotis.seq), nbchar
= 80, file.out = "myotis_µsat.fas")
>
```

Locus Name	GenBank no.	Temp. (°C)	Primer [µM]	$MgCl_2$ [µM]	Primer sequence (5′–3′)	Repeat motif	Fragment size (bp)
A13	AF203638	55	0.45	2.0	AACGTTCATTCTGCCAAAGG TCATGCTGTTCCACTTCTGG	$(TC)_5TT(TC)_{25}$	237
B11	AF203644	55	0.25	1.5	CATTTACAACGTTACTCCAGGC GCAGTTAATCTGCATTTTGCC	$(TG)_6TA(TG)_{15}$	224
B22	AF203645	55	0.35	2.0	CCAGAAACACCAACGTCATG GGAGGGGTTGTACAGAGAAGG	$(TC)_{22}$	230
C113	AF203647	60	0.25	1.5	ACCTCCCTGCCCTGCAC GCAATGCTTCCTCCAAGTCC	$(ACC)_7$	101
D9	AF203649	60	0.25	2.0	TCTTTCCTCCCCTGTGCTC TCTGGACCCAAAATGCAGG	$(CT)_{29}$	148
D15	AF203651	60	0.45	2.0	GCTCTCTGAAGAGGCCCTG ATTCCAAGAGTGACAGCATCC	$(AC)_{17}$	114
E24	AF203657	60	0.25	1.5	GCAGGTTCAATCCCTGACC AAAGCCAGACTCCAAATTCTG	$(TC)_{32}$	236
F19	AF203661	60	0.25	1.5	GCTAGCCATGGAGAAGGAAG CCCAAATCTGTCTTTCAGGC	$(CA)_{19}$	214
G9	AF203666	55	0.25	1.5	AGGGGACATACAAGAATCAACC TAATTTCTCCACTGAACTCCCC	$(TC)_{19}$	153
G25	AF203667	55	0.25	1.5	TCCTTCCCATTTCTGTGAGG CCATTTCATCCATCCAGTCC	$(AGC)_{11}AAT(AGC)_4$	147
G30	AF203669	55	0.25	1.5	TTGCCAAATTCTGGTATCTTCC AGAGCTTAATGGGGAGGCTG	$(AG)_{24}$	132
H19	AF203675	55	0.25	1.5	GGAATCCGAATCCCTGGC GACATCCCCTCACCCCAAC	$(GT)_{18}CT(GT)_2$	100
H29	AF203677	55	0.35	1.5	TCAGGTGAGGATTGAAAACAC GCTTTATTTAGCATTGGAGAGC	$(CA)_{21}$	182

Tabella E6.2 Characteristics and PCR conditions for the 13 microsatellite loci developed in the bat *Myotis myotis*. Temp. is the annealing temperature and Primer the final concentration of each primer. Repeat motifs and fragment sizes are those of the cloned allele.

>AF203638 **A13 locus**
GCCACACTCTGAGCCTAGAAGGTGTGACACATGACCTTCAGGAAAGGTGACATCTCATTGGGGAAACTGTCAATGAAACA
GTGAAATCCCAAGGTAAGAGCCTAAAAGTGTGGC**AACGTTCATTCTGCCAAAGG**AATTCAGTAAAGGATGAGAGATGAGC
GCATGCTTGGAGGGGTCTGAAAAGTTTTCACAATGGTGGGGTAGGTGTCTAACTGCTGGGGTCAGAGGGTTGGGGAAAAG
GGTGAGGGGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAGAAAGAGAGAGAGAGGGAAATGAA
TGTCTAGCTTTCCAGAAGTGGAACAGCATGAACGAAATTCCCAAGGCTTGGAGAGCAAGGCTGGTGGGAAAGCAGCAGCT
GACCCGTTGGAACAGTCAGGAGGCAGCTGGGTGTGAAATCCCAGGAACTCTGATTCCCCAGGCTTGGGTCCTTCCTTAAA
ATTTTCCTAACGGTGCAGTCAAGCCCAGGACAAAGTTGTTTTCTTATCCAGTGTGTGATC

>AF203644 **B11 locus**
gatcagaaaatgtttctcattctctctttttttaaattttaaagtttcatgcttgaatttgtggactaatgttccattaa
agtatac**catttacaacgttactccaggc**aaaaatttttagtatgcattatttagattgtgagcatatacacttcattgt
gtgtgtgtgtatgtgtgtgtgtgtgtgtgtgtgtgtgtgtgtttggaacttagcaccaacaaatcttatctaataaaaga
gaaacatagtaattagcgtacaaccgctacccttcccattggctaatcagggcaaaatgcagattaactgccagccaaga
tggcggccggcagccaggcagctggaagtgaacatgaggcttgcttgcttcagtgagggagaactccaatgttccccgcc
tgcctttgccggcctctgagctacaactctaagcaactatgttacaaatatagaagctaaacaaaaccccagaaacctgc
tttagtcgcagggcttcatccagcaggatc

...

Forward primer
Reverse primer?

```
> install.packages('spgs')
> library(spgs)
> complement('catttacaacgttactccaggc')
[1] "gtaaatgttgcaatgaggtccg"
>
```

PCR reactions were carried out in 7.5 µL reaction volumes, with 2.5 µL of template DNA (25 – 100 ng), 0.1 mM of each dNTP, 1 × Taq buffer and 0.25 U of Taq DNA polymerase (Qiagen). One primer of each pair was fluorescent labelled (from Applied Biosystems) with HEX (locus A13, D15, G25), NED (locus B11, B22, G30, H29) or 6-FAM (locus C113, D9, E24, F19, G9, H19). PCR reactions were overlaid with mineral oil and performed with a PTC-100 thermocycler (MJ research) as follows: 3 min at 95 °C followed by 30 cycles of 45 s at 94 °C, 30 s at the annealing temperature (Table 2.2), 45 s at 72 °C. The cycles ended with one final extension of 10 min at 72 °C (Table E6.3).

	Castella e Ruedi (2000)	Castella *et al.* (2001)
Marcatore genetico	µsatelliti	mtDNA
Volume della master mix	7.5 µl	50 µl
Acqua DNA-free		100 µl
KCl		
Tris-HCl		
$MgCl_2$	1.5÷2.0 µM	2.5 mM
primer	0.25÷0.45 µM	0.2 µM
forward		
reverse		
dNTP	0.1 mM	200 µM
Tipo di polimerasi	Taq	Taq DNA Polymerase 1 (Qiagen)
Quantità di enzima	0.25 unità	2 unità
Quantità di DNA	2.5 µl (25÷100 ng)	10 µl
Temperatura di attivazione della polim.	95°C	95°C
Tempo di attivazione della polim.	3 minuti	3 minuti
Temperatura di denaturazione	94 °C	94 °C
Tempo di denaturazione	45 secondi	45 secondi
Temperatura di *annealing* del *primer*	55÷60°C	50°C
Tempo di *annealing* del *primer*	30 secondi	45 secondi
Temperatura di estensione del *primer*	72°C	72°C
Tempo di estensione del *primer*	45 secondi	1 minuto
N° di cicli	30	35
Tempo di estensione finale	10 minuti	7 minuti

Tabella E6.3 Parametri utilizzati per la PCR in due studi diversi.

We designed two post-PCR multiplex combinations to simultaneously reveal in a single lane six and seven loci, respectively. Prior to electrophoresis, the PCR products of each individual were mixed in the following proportions. The first multiplex contained 1.25 µL of C113, 3.0 µL of D9 and 2.5 µL of each B11, E24, G25 and H29. For the second set of *loci*, 3 µL of each B22, G9 and G30 were mixed with 2 µL of each F19 and H19, and 5 µL of each A13 and D15. PCR products were run on an ABI 373XL sequencer and sized with internal lane standard (GenScan 350-Rox; Applied Biosystems) using the program genescan version 3.1 (Applied Biosystems) and by comparison with the comigrating individual that was used to construct the clone library.

E6.3 Il DNA mitocondriale (mtDNA)
(Castella *et al.*, 2001)

Genomic DNA was extracted from half wing punches following a salt/chloroform procedure modified from Miller *et al.* (1988) by adding one step of chloroform/isoamyl alcohol (24/1) extraction to the original protocol. The precipitated DNA was resuspended in 100 µl sterile water. The mitochondrial control region (CR) is a noncoding segment located between the genes for tRNAPhe and the tRNAPro. Although this region is highly variable at the sequence level, this variation is mainly confined to two hypervariable domains (HVI and HVII) on the 5' and 3' sides of the CR (Vigilant *et al*., 1991).
In our study, the HVII was PCR-amplified using the primers L16517 (Fumagalli *et al.*, 1996) and sH651, a shorter version of the primer H00651 (Kocher *et al.*, 1989). The sequence for this redesigned primer was 5'-**AAGGCTAGGACCAAACCT**-3'. Amplifications were carried out in 50 µl reaction volumes overlaid with mineral oil. The following conditions were used: 10 µl of template DNA, 2.5 mM $MgCl_2$, 0.2 µM of each primer, 200 µM of each dNTP, 2 units of Taq DNA Polymerase 1 (Qiagen) with its buffer and the Q-solution. Thermal profiles started with an initial denaturation at 95 °C for 3 min, followed by 35 cycles consisting of 45 s at 94 °C, 45 s at 50 °C, 1 min at 72 °C. The cycles ended with one final extension of 7 min at 72 °C. PCR products were purified with QIAquick spin columns (Qiagen) following the manufacturer's instructions.

Four microlitres (10±30 ng) of cleaned PCR products were then mixed with 4 µl of Dye Terminator Reaction Kit 2 (Applied Biosystems) and 0.5 µM of primer L16517 in a total volume of 10 µl. Thermal profiles for the sequencing reaction consisted of 25 cycles of 30 s at 96 °C, 20 s at 50 °C and 4 min at 60 °C. The sequences were run on 5% Long Ranger (BioConcept), 6 M urea gels in an ABI 373XL automated sequencer (Applied Biosystems).
The same 260 bats were typed at the following 15 microsatellite *loci*: A13, B11, B22, C113, D9, D15, E24, F19, G9, G25, G30, H19 and H29, as previously described in Castella & Ruedi (2000), and MM5' and NN8', that are redesigned versions of MM5 and NN8 of Petri et al. (1997). Modified primer sequences for these last two *loci* are 5'-AGGGTTGATTTAACATGC-3' and 5'-CTTTCATCCAGTTCTGG-3' for MM5' and 5'-GGTGATTTCCATTCCCAGC-3' and 5'-TTGTGTTTTAAAGAAAATCCG-3' for NN8'. PCR amplifications for these two additional *loci* were carried out in 10 µl reaction volumes. For both of them, we used 3 µl of template DNA (5±10 ng), 1.5 mM MgCl2, 0.5 µM of each primer, 80 lM dNTP (dATP concentration 1:10), 0.02 µl (^{33}P)-dATP at 1000 Ci mmol^-1, 0.3 U Taq DNA polymerase (Qiagen) and its buffer. PCR reactions were overlaid with mineral oil and performed with a PTC-100 thermocycler 4 (MJ research) as follows: 2 min at 95 °C followed by 35 cycles of 45 s at 94 °C, 45 s at 50 °C, 45 s at 72 °C. The cycles ended with one final extension of 10 min at 72 °C. Samples were electrophoresed on denaturing polyacrylamide gels (6%, 8 M urea).